

Curve Fitting/ Interpolation

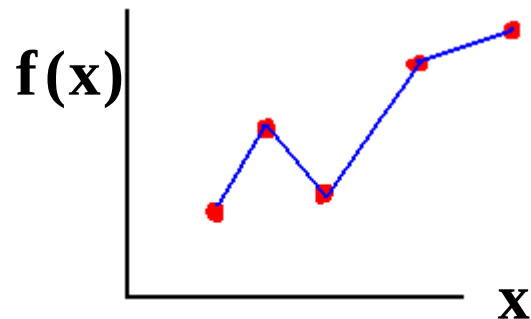
By

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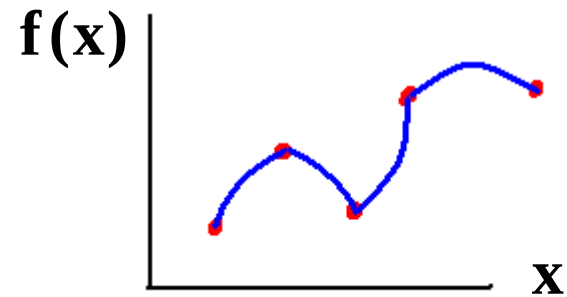
Interpolation

Interpolation: is to fit a straight lines or curves that pass directly through each of the points.

Remember that the least square regression is to derive a single straight line or curve to represent general trend of the data, we make no effort to interest every point



Linear Interpolation



Curvilinear Interpolation

Question: what is the value of y at $x=2.5$?

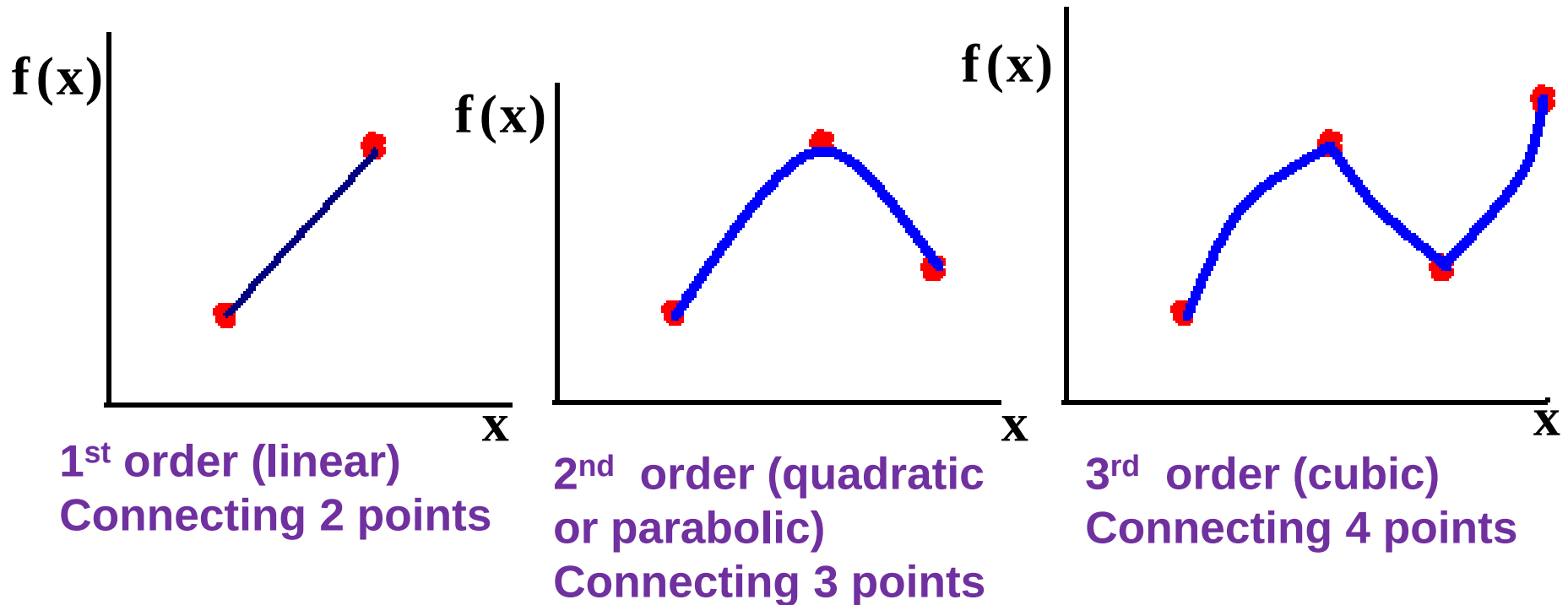
x	1	2	3	4
y	6	8	11	15

Polynomial Interpolation:

The most common method is polynomial interpolation:

$$f(x) = a_0 + a_1x + a_2x^2 + \dots + a_nx^n$$

nth-order polynomial
(n+1) data points



Forms of Interpolating Polynomial

(1) Newton's Divided-Difference

- Linear Interpolation (1st order)
- Quadratic Interpolation (2nd order)
- General form Polynomials

(2) Lagrange

Newton's Divided-Difference Interpolating Polynomials

Linear Interpolation (1st order Polynomial)

$$f_1(x) = b_0 + b_1(x - x_0) \quad (1)$$

Linear interpolation is to connect 2 data points with a straight line

Using Similar Triangles

$$\frac{f_1(x) - f(x_0)}{x - x_0} = \frac{f(x_1) - f(x_0)}{x_1 - x_0}$$

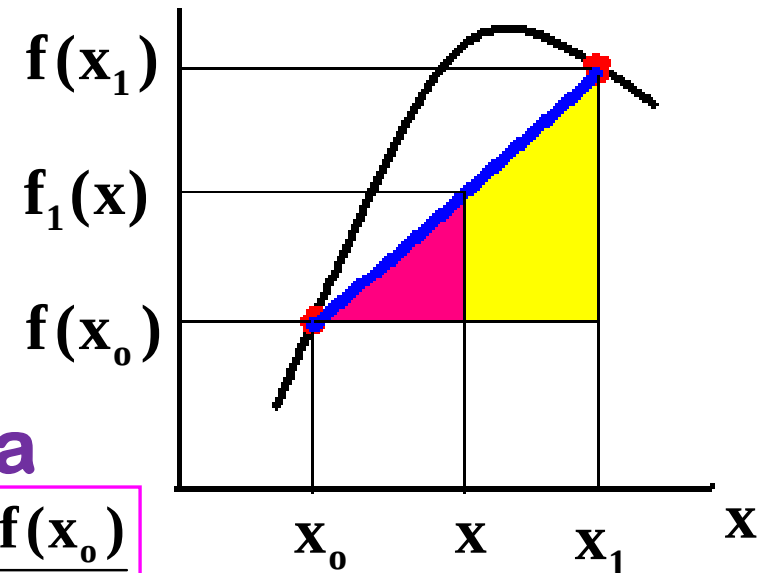
$$f_1(x) = f(x_0) + \frac{f(x_1) - f(x_0)}{x_1 - x_0}(x - x_0)$$

Linear Interpolating Formula

Compare with (1):

$$b_0 = f(x_0)$$

$$b_1 = \frac{f(x_1) - f(x_0)}{x_1 - x_0}$$



Example: Estimate $\text{Ln}(2)$ using linear interpolation for interval $[1 \ 4]$

Solution

$$f_1(x) = f(x_0) + \frac{f(x_1) - f(x_0)}{x_1 - x_0}(x - x_0)$$

$$\begin{aligned} f_1(x) &= f(1) + \frac{f(4) - f(1)}{4 - 1}(x - 1) = \text{Ln}(1) + \frac{\text{Ln}(4) - \text{Ln}(1)}{4 - 1}(x - 1) \\ &= 0.462098(x - 1) \end{aligned}$$

$$f_1(x) = 0.462098(x - 1)$$

At $x=2$, then

$$f_1(2) = 0.462098$$

True value:
 $\text{Ln}(2) = 0.693147$

$$\varepsilon_t = \left| \frac{0.693147 - 0.462098}{0.693147} \right| 100\% = 33.3\%$$

Example: Estimate $f(4.5)$ using linear interpolation from the given table.

x	3	5
f(x)	19	35

Solution

$$f_1(x) = f(x_0) + \frac{f(x_1) - f(x_0)}{x_1 - x_0} (x - x_0)$$

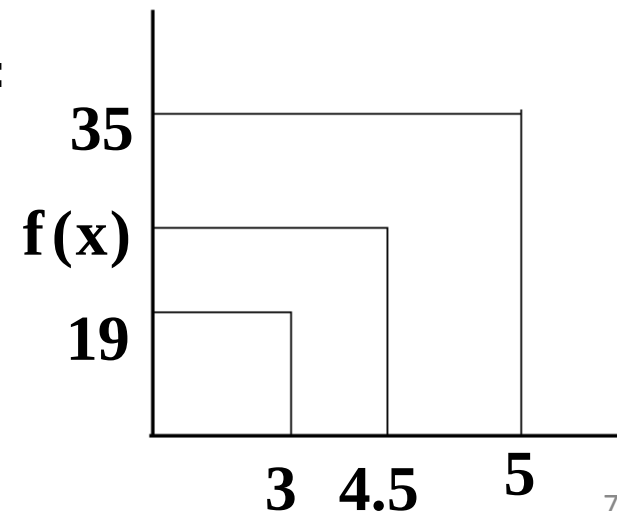
$$\begin{aligned} f_1(4.5) &= f(3) + \frac{f(5) - f(3)}{5 - 3} (4.5 - 3) \\ &= 19 + \frac{35 - 19}{5 - 3} (4.5 - 3) \\ &= 31 \end{aligned}$$

Alternative Solution

Do the following if linear interpolation is needed:

$$\frac{35 - 19}{5 - 3} = \frac{35 - f(4.5)}{5 - 4.5} \Rightarrow f(4.5) = 31$$

Question: what will be x at $f(x)=22$?



Quadratic Interpolation (2nd order Polynomial)

“Parabolic”

2nd order (quadratic or parabolic) is connecting 3 points

$$f_2(x) = b_0 + b_1(x - x_0) + b_2(x - x_0)(x - x_1) \quad (1)$$

Procedures to determine the value of coefficients:

For b_0 , Let $x=x_0 \Rightarrow b_0 = f(x_0)$ See linear interpolation

For b_1 , Let $x=x_1 \Rightarrow b_1 = \frac{f(x_1) - f(x_0)}{x_1 - x_0}$ See linear interpolation

For b_2 , Let $x=x_2 \Rightarrow b_2 = \frac{\frac{f(x_2) - f(x_1)}{x_2 - x_1} - \frac{f(x_1) - f(x_0)}{x_1 - x_0}}{x_2 - x_0}$

Example: Estimate $\text{Ln}(2)$ using 2nd order polynomial to 3 points [1 4 6]

Solution

$$b_0 = f(x_0)$$

$$b_0 = f(1) = \text{Ln}(1) = 0$$

$$b_1 = \frac{f(x_1) - f(x_0)}{x_1 - x_0}$$

$$b_1 = \frac{f(4) - f(1)}{4 - 1} = \frac{\text{Ln}(4) - \text{Ln}(1)}{4 - 1} = 0.4620981$$

$$b_2 = \frac{\frac{f(x_2) - f(x_1)}{x_2 - x_1} - \frac{f(x_1) - f(x_0)}{x_1 - x_0}}{x_2 - x_0}$$

$$b_2 = \frac{\frac{\text{Ln}(6) - \text{Ln}(4)}{6 - 4} - \frac{\text{Ln}(4) - \text{Ln}(1)}{4 - 1}}{6 - 1} = -0.0518731$$

$$f_2(x) = b_0 + b_1(x - x_0) + b_2(x - x_0)(x - x_1)$$

$$f_2(x) = \text{Ln}(x) = 0 + 0.4620981(x - 1) + (-0.0518731)(x - 1)(x - 4)$$

$$f_2(x) = 0.4620981(x - 1) + (-0.0518731)(x - 1)(x - 4)$$

At $x=2$, then

$$f_2(2) = 0.5658443$$

General Form of Interpolating Polynomial

$$f_n(x) = b_0 + b_1(x - x_0) + b_2(x - x_0)(x - x_1) + b_3(x - x_0)(x - x_1)(x - x_2) + \dots + b_n(x - x_0)\dots(x - x_{n-1}) \quad (1)$$

$$b_0 = f(x_0)$$

$$b_1 = f[x_1, x_0]$$

$$b_2 = f[x_2, x_1, x_0]$$

$$b_3 = f[x_3, x_2, x_1, x_0]$$

$$b_n = f[x_n, x_{n-1}, \dots, x_1, x_0]$$

Where the bracketed function evaluations are:

FINITE DIVIDED DIFFERENCES

Examples:

(1) The **1st finite divided difference** is represented generally as

$$f[x_i, x_j] = \frac{f(x_i) - f(x_j)}{x_i - x_j}$$

$$b_1 = f[x_1, x_0] = \frac{f(x_1) - f(x_0)}{x_1 - x_0}$$

(2) The **2nd finite divided difference** is represented generally as

$$f[x_i, x_j, x_k] = \frac{f[x_i, x_j] - f[x_j, x_k]}{x_i - x_k}$$

$$b_2 = f[x_2, x_1, x_0] = \frac{f[x_2, x_1] - f[x_1, x_0]}{x_2 - x_0} = \frac{\frac{f(x_2) - f(x_1)}{x_2 - x_1} - \frac{f(x_1) - f(x_0)}{x_1 - x_0}}{x_2 - x_0}$$

(3) The **3rd finite divided difference** is represented generally as

$$\begin{aligned} f[x_i, x_j, x_k, x_L] &= \frac{f[x_i, x_j, x_k] - f[x_j, x_k, x_L]}{x_i - x_L} = \frac{\frac{f[x_i, x_j] - f[x_j, x_k]}{x_i - x_k} - \frac{f[x_j, x_k] - f[x_k, x_L]}{x_j - x_L}}{x_i - x_L} \\ &= \frac{\frac{\frac{f(x_i) - f(x_j)}{x_i - x_j} - \frac{f(x_j) - f(x_k)}{x_j - x_k}}{x_i - x_k} - \frac{\frac{f(x_j) - f(x_k)}{x_j - x_k} - \frac{f(x_k) - f(x_L)}{x_k - x_L}}{x_j - x_L}}{x_i - x_L} \end{aligned}$$

$$\begin{aligned}
 b_3 = f[x_3, x_2, x_1, x_0] &= \frac{f[x_3, x_2, x_1] - f[x_2, x_1, x_0]}{x_3 - x_0} = \\
 &= \frac{\frac{f[x_3, x_2] - f[x_2, x_1]}{x_3 - x_1} - \frac{f[x_2, x_1] - f[x_1, x_0]}{x_2 - x_0}}{x_3 - x_0} = \\
 &= \frac{\frac{f(x_3) - f(x_2)}{x_3 - x_2} - \frac{f(x_2) - f(x_1)}{x_2 - x_1}}{x_3 - x_1} - \frac{\frac{f(x_2) - f(x_1)}{x_2 - x_1} - \frac{f(x_1) - f(x_0)}{x_1 - x_0}}{x_2 - x_0}
 \end{aligned}$$

(4) The **n^{th} finite divided difference** is represented generally as

$$f[x_n, x_{n-1}, \dots, x_1, x_0] = \frac{f[x_n, x_{n-1}, \dots, x_1] - f[x_{n-1}, x_{n-2}, \dots, x_0]}{x_n - x_0}$$

$$f_n(x) = b_0 + b_1(x - x_0) + b_2(x - x_0)(x - x_1) + b_3(x - x_0)(x - x_1)(x - x_2) + \dots + b_n(x - x_0)\dots(x - x_{n-1}) \quad (1)$$

$$f_n(x) = f(x_0) + (x - x_0)f[x_1, x_0] + (x - x_0)(x - x_1)f[x_2, x_1, x_0] + (x - x_0)(x - x_1)(x - x_2)f[x_3, x_2, x_1, x_0] + \dots + (x - x_0)(x - x_1)\dots(x - x_{n-1})f[x_n, x_{n-1}, \dots, x_0]$$



Newton's divided-difference
interpolating polynomial

Example: Estimate $\text{Ln}(2)$ using 3rd order polynomial (cubic) to 4 points [1 4 6 5]

Solution

$$\begin{array}{cccc} \downarrow & \downarrow & \downarrow & \downarrow \\ [x_0 & x_1 & x_2 & x_3] \end{array}$$

$$f_3(x) = b_0 + b_1(x - x_0) + b_2(x - x_0)(x - x_1) + b_3(x - x_0)(x - x_1)(x - x_2)$$

$$b_0 = f(x_0)$$

$$b_0 = \text{Ln}(1) = 0$$

$$b_1 = f[x_1, x_0] = \frac{f(x_1) - f(x_0)}{x_1 - x_0}$$

$$b_1 = \frac{\text{Ln}(4) - f(1)}{4 - 1} = 0.4620981$$

$$b_2 = f[x_2, x_1, x_0] = \frac{f[x_2, x_1] - f[x_1, x_0]}{x_2 - x_0} = \frac{\frac{f(x_2) - f(x_1)}{x_2 - x_1} - \frac{f(x_1) - f(x_0)}{x_1 - x_0}}{x_2 - x_0}$$

$$b_2 = \frac{\frac{\text{Ln}(6) - \text{Ln}(4)}{6 - 4} - \frac{\text{Ln}(4) - \text{Ln}(1)}{4 - 1}}{6 - 1} = -0.0518732$$

$$\begin{aligned}
 b_3 = f[x_3, x_2, x_1, x_0] &= \frac{f[x_3, x_2, x_1] - f[x_2, x_1, x_0]}{x_3 - x_0} = \\
 &= \frac{\frac{f[x_3, x_2] - f[x_2, x_1]}{x_3 - x_1} - \frac{f[x_2, x_1] - f[x_1, x_0]}{x_2 - x_0}}{x_3 - x_0} = \\
 &= \frac{\frac{f(x_3) - f(x_2)}{x_3 - x_2} - \frac{f(x_2) - f(x_1)}{x_2 - x_1}}{x_3 - x_1} - \frac{\frac{f(x_2) - f(x_1)}{x_2 - x_1} - \frac{f(x_1) - f(x_0)}{x_1 - x_0}}{x_2 - x_0}
 \end{aligned}$$

$$b_3 = \frac{\frac{\ln(5) - \ln(6)}{5 - 6} - \frac{\ln(6) - \ln(4)}{6 - 4}}{5 - 4} - \frac{\frac{\ln(6) - \ln(4)}{6 - 4} - \frac{\ln(4) - \ln(1)}{4 - 1}}{6 - 1} = 0.00786555075$$

$$f_3(x) = 0.4620981(x - 1) - 0.0518732(x - 1)(x - 4) + 0.00786555075(x - 1)(x - 4)(x - 6)$$

At x=2

$$f_3(2) = 0.6287686$$

Example: Given the following data. Calculate $f(4)$ using Newton's interpolating polynomials of order 1 through 4.

x	3	5	2	7	1
f(x)	19	99	6	291	3

Solution

$$f_4(x) = b_0 + b_1(x - x_0) + b_2(x - x_0)(x - x_1) + b_3(x - x_0)(x - x_1)(x - x_2) + b_4(x - x_0)(x - x_1)(x - x_2)(x - x_3)$$

$$b_0 = f(x_0)$$

$$b_1 = f[x_1, x_0]$$

$$b_2 = f[x_2, x_1, x_0]$$

$$b_3 = f[x_3, x_2, x_1, x_0]$$

$$b_4 = f[x_4, x_3, x_2, x_1, x_0]$$

i	x_i	$f(x_i)$	$f[x_{i+1}, x_i]$	$f[x_{i+2}, x_{i+1}, x_i]$	$f[x_{i+3}, x_{i+2}, x_{i+1}, x_i]$	$f[x_{i+4}, x_{i+3}, x_{i+2}, x_{i+1}, x_i]$
0	3	19= b_0	$f[x_1, x_0]$ =40= b_1	$f[x_2, x_1, x_0]=$ $\frac{f[x_2, x_1]-f[x_1, x_0]}{x_2-x_0}$ =9= b_2	$f[x_3, x_2, x_1, x_0]=$ $\frac{f[x_3, x_2, x_1]-f[x_2, x_1, x_0]}{x_3-x_0}$ =1= b_3	$f[x_4, x_3, x_2, x_1, x_0]=$ $\frac{f[x_4, x_3, x_2, x_1]-f[x_3, x_2, x_1, x_0]}{x_4-x_0}$ =0= b_4
1	5	99	$f[x_2, x_1]=31$	$f[x_3, x_2, x_1]=$ $\frac{f[x_3, x_2]-f[x_2, x_1]}{x_3-x_1}$ =13	$f[x_4, x_3, x_2, x_1]=$ $\frac{f[x_4, x_3, x_2]-f[x_3, x_2, x_1]}{x_4-x_1}$ =1	
2	2	6	$f[x_3, x_2]=57$	$f[x_4, x_3, x_2]=$ $\frac{f[x_4, x_3]-f[x_3, x_2]}{x_4-x_2}$ =9		
3	7	291	$f[x_4, x_3]=48$			
4	1	3				

$$\begin{aligned}f_1(4) &= b_0 + b_1(x - x_0) \\ &= 19 + 40(4 - 3) = 59\end{aligned}$$

$$\begin{aligned}f_2(4) &= f_1(4) + b_2(x - x_0)(x - x_1) \\ &= 59 + 9(4 - 3)(4 - 5) = 50\end{aligned}$$

$$\begin{aligned}f_3(4) &= f_2(4) + b_3(x - x_0)(x - x_1)(x - x_2) \\ &= 50 + 1(4 - 3)(4 - 5)(4 - 2) = 48\end{aligned}$$

$$\begin{aligned}f_4(4) &= f_3(4) + b_4(x - x_0)(x - x_1)(x - x_2)(x - x_3) \\ &= 48 + 0(4 - 3)(4 - 5)(4 - 2)(4 - 7) = 48\end{aligned}$$

Example: Use quadratic polynomial of Newton's divided difference interpolating to find x of $f(x)=0.3$ for the following data

x	2	3	4
f(x)	0.5	0.3333	0.25

Solution

$$f_2(x) = b_0 + b_1(x - x_0) + b_2(x - x_0)(x - x_1) \quad (1)$$

$$b_0 = f(x_0)$$

$$b_0 = f(2) = 0.5$$

We are talking about **inverse interpolation**

$$b_1 = \frac{f(x_1) - f(x_0)}{x_1 - x_0}$$

$$b_1 = \frac{0.3333 - 0.5}{3 - 2} = -0.1667$$

$$b_2 = \frac{\frac{f(x_2) - f(x_1)}{x_2 - x_1} - \frac{f(x_1) - f(x_0)}{x_1 - x_0}}{x_2 - x_0}$$

$$b_2 = \frac{\frac{0.25 - 0.3333}{4 - 3} - \frac{0.3333 - 0.5}{3 - 2}}{4 - 2} = 0.0417$$

$$f_2(x) = 0.5 - 0.1667(x - 2) + 0.0417(x - 2)(x - 3) = 0.3$$

$$x = 5.704158, 3.295842$$

$$x = 3.295842$$

Lagrange Interpolating Polynomials

It is derived from Newton's Polynomials

$$f_n(x) = \sum_{i=0}^n L_i(x) f(x_i)$$

$$L_i(x) = \prod_{\substack{j=0 \\ j \neq i}}^n \frac{x - x_j}{x_i - x_j}$$

$\prod$: Product of

Examples:

(1) 1st order Lagrange Polynomial, $n=1$ (Linear version)

$$f_1(x) = \sum_{i=0}^1 L_i(x) f(x_i) = L_0(x) f(x_0) + L_1(x) f(x_1)$$

$$L_0(x) = \prod_{\substack{j=0 \\ j \neq i \rightarrow j \neq 0}}^1 \frac{x - x_j}{x_0 - x_j} = \frac{x - x_1}{x_0 - x_1}$$

$$L_1(x) = \prod_{\substack{j=0 \\ j \neq i \rightarrow j \neq 1}}^1 \frac{x - x_j}{x_1 - x_j} = \frac{x - x_0}{x_1 - x_0}$$

$$f_1(x) = \frac{x - x_1}{x_0 - x_1} f(x_0) + \frac{x - x_0}{x_1 - x_0} f(x_1)$$

(2) 2nd order Lagrange Polynomial, n=2

$$f_2(x) = \sum_{i=0}^2 L_i(x) f(x_i) = L_0(x) f(x_0) + L_1(x) f(x_1) + L_2(x) f(x_2)$$

$$L_0(x) = \prod_{\substack{j=0 \\ j \neq i \rightarrow j \neq 0}}^2 \frac{x - x_j}{x_0 - x_j} = \frac{x - x_1}{x_0 - x_1} \frac{x - x_2}{x_0 - x_2}$$

$$L_1(x) = \prod_{\substack{j=0 \\ j \neq i \rightarrow j \neq 1}}^2 \frac{x - x_j}{x_1 - x_j} = \frac{x - x_0}{x_1 - x_0} \frac{x - x_2}{x_1 - x_2}$$

$$L_2(x) = \prod_{\substack{j=0 \\ j \neq i \rightarrow j \neq 2}}^2 \frac{x - x_j}{x_2 - x_j} = \frac{x - x_0}{x_2 - x_0} \frac{x - x_1}{x_2 - x_1}$$

$$f_2(x) = \frac{x - x_1}{x_0 - x_1} \frac{x - x_2}{x_0 - x_2} f(x_0) + \frac{x - x_0}{x_1 - x_0} \frac{x - x_2}{x_1 - x_2} f(x_1) + \frac{x - x_0}{x_2 - x_0} \frac{x - x_1}{x_2 - x_1} f(x_2)$$

(3) 3rd order Lagrange Polynomial, n=3

$$f_3(x) = \sum_{i=0}^3 L_i(x)f(x_i) = L_0(x)f(x_0) + L_1(x)f(x_1) + L_2(x)f(x_2) + L_3(x)f(x_3)$$

$$L_0(x) = \prod_{\substack{j=0 \\ j \neq i \rightarrow j \neq 0}}^3 \frac{x - x_j}{x_0 - x_j} = \frac{x - x_1}{x_0 - x_1} \frac{x - x_2}{x_0 - x_2} \frac{x - x_3}{x_0 - x_3}$$

$$L_1(x) = \prod_{\substack{j=0 \\ j \neq i \rightarrow j \neq 1}}^3 \frac{x - x_j}{x_1 - x_j} = \frac{x - x_0}{x_1 - x_0} \frac{x - x_2}{x_1 - x_2} \frac{x - x_3}{x_1 - x_3}$$

$$L_2(x) = \prod_{\substack{j=0 \\ j \neq i \rightarrow j \neq 2}}^3 \frac{x - x_j}{x_2 - x_j} = \frac{x - x_0}{x_2 - x_0} \frac{x - x_1}{x_2 - x_1} \frac{x - x_3}{x_2 - x_3}$$

$$L_3(x) = \prod_{\substack{j=0 \\ j \neq i \rightarrow j \neq 3}}^3 \frac{x - x_j}{x_3 - x_j} = \frac{x - x_0}{x_3 - x_0} \frac{x - x_1}{x_3 - x_1} \frac{x - x_2}{x_3 - x_2}$$

$$\begin{aligned} f_3(x) = & \frac{x - x_1}{x_0 - x_1} \frac{x - x_2}{x_0 - x_2} \frac{x - x_3}{x_0 - x_3} f(x_0) + \\ & \frac{x - x_0}{x_1 - x_0} \frac{x - x_2}{x_1 - x_2} \frac{x - x_3}{x_1 - x_3} f(x_1) + \\ & \frac{x - x_0}{x_2 - x_0} \frac{x - x_1}{x_2 - x_1} \frac{x - x_3}{x_2 - x_3} f(x_2) + \\ & \frac{x - x_0}{x_3 - x_0} \frac{x - x_1}{x_3 - x_1} \frac{x - x_2}{x_3 - x_2} f(x_3) \end{aligned}$$

Example: Use Lagrange interpolating polynomial of the 1st order to evaluate $\text{Ln}(2)$ to two points [1 4]

Solution

$$x_0 = 1$$

$$x_1 = 4$$

$$f_1(x) = \frac{x - x_1}{x_0 - x_1} f(x_0) + \frac{x - x_0}{x_1 - x_0} f(x_1)$$

$$f_1(x) = \frac{x - 4}{1 - 4} \text{Ln}(1) + \frac{x - 1}{4 - 1} \text{Ln}(4)$$

At $x=2$

$$f_1(2) = \frac{2 - 4}{1 - 4} \text{Ln}(1) + \frac{2 - 1}{4 - 1} \text{Ln}(4) = 0.4620981$$

Example: Use Lagrange interpolating polynomial of the 2nd order to evaluate **Ln(2)** to three points [1 4 6]

Solution

$$x_0 = 1, \quad x_1 = 4, \quad x_2 = 6$$

$$f_2(x) = \frac{x-x_1}{x_0-x_1} \frac{x-x_2}{x_0-x_2} f(x_0) + \frac{x-x_0}{x_1-x_0} \frac{x-x_2}{x_1-x_2} f(x_1) + \frac{x-x_0}{x_2-x_0} \frac{x-x_1}{x_2-x_1} f(x_2)$$

$$f_2(x) = \frac{x-4}{1-4} \frac{x-6}{1-6} \text{Ln}(1) + \frac{x-1}{4-1} \frac{x-6}{4-6} \text{Ln}(4) + \frac{x-1}{6-1} \frac{x-4}{6-4} \text{Ln}(6)$$

At x=2

$$f_2(2) = \frac{2-4}{1-4} \frac{2-6}{1-6} \text{Ln}(1) + \frac{2-1}{4-1} \frac{2-6}{4-6} \text{Ln}(4) + \frac{2-1}{6-1} \frac{2-4}{6-4} \text{Ln}(6) = 0.5658444$$

Example: Evaluate $f(4)$ using Lagrange interpolating polynomial of order 1 through 3

x	3	5	2	7	1
f(x)	19	99	6	291	3

Solution

$$x_0 = 3, \quad x_1 = 5, \quad x_2 = 2, \quad x_3 = 7, \quad x_4 = 1$$

1st order

$$f_1(x) = \frac{x - x_1}{x_0 - x_1} f(x_0) + \frac{x - x_0}{x_1 - x_0} f(x_1)$$

$$f_1(4) = \frac{4 - 5}{3 - 5} (19) + \frac{4 - 3}{5 - 3} (99) = 59$$

2nd order

$$f_2(x) = \frac{x-x_1}{x_0-x_1} \frac{x-x_2}{x_0-x_2} f(x_0) + \frac{x-x_0}{x_1-x_0} \frac{x-x_2}{x_1-x_2} f(x_1) + \frac{x-x_0}{x_2-x_0} \frac{x-x_1}{x_2-x_1} f(x_2)$$

$$f_2(4) = \frac{4-5}{3-5} \frac{4-2}{3-2} (19) + \frac{4-3}{5-3} \frac{4-2}{5-2} (99) + \frac{4-3}{2-3} \frac{4-5}{2-5} (6) = 50$$

3rd order

$$\begin{aligned} f_3(x) = & \frac{x-x_1}{x_0-x_1} \frac{x-x_2}{x_0-x_2} \frac{x-x_3}{x_0-x_3} f(x_0) + \\ & \frac{x-x_0}{x_1-x_0} \frac{x-x_2}{x_1-x_2} \frac{x-x_3}{x_1-x_3} f(x_1) + \\ & \frac{x-x_0}{x_2-x_0} \frac{x-x_1}{x_2-x_1} \frac{x-x_3}{x_2-x_3} f(x_2) + \\ & \frac{x-x_0}{x_3-x_0} \frac{x-x_1}{x_3-x_1} \frac{x-x_2}{x_3-x_2} f(x_3) \end{aligned}$$

$$\begin{aligned} f_3(4) = & \frac{4-5}{3-5} \frac{4-2}{3-2} \frac{4-7}{3-7} (19) + \\ & \frac{4-3}{5-3} \frac{4-2}{5-2} \frac{4-7}{5-7} (99) + \\ & \frac{4-3}{2-3} \frac{4-5}{2-5} \frac{4-7}{2-7} (6) + \\ & \frac{4-3}{7-3} \frac{4-5}{7-5} \frac{4-2}{7-2} (291) \\ = & 48 \end{aligned}$$

Question for You!

Evaluate $f(4)$ using Lagrange interpolating polynomial of order 1 through 3

x	-2	1	3	5	6	10	13	20
f(x)	1.5	10	19	99	6	291	3	32